

A Possible Drug-Drug Interaction Between Eliquis and Amiodarone Resulting in Hemopericardium

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Case presentation

An 80-year-old male with a past medical history of paroxysmal atrial fibrillation, heart failure with preserved ejection fraction, chronic obstructive pulmonary disease, tobacco dependence, benign essential hypertension, and stage 3 chronic kidney disease presented to the emergency department (ED) with shortness of breath, orthopnea, increasing lower extremity edema, and cough of two weeks' duration. The cough was productive with white sputum and worse in the supine position. He denied hemoptysis, chest pain, fever, chills, night sweat, weight loss, nausea, and vomiting. His paroxysmal atrial fibrillation was diagnosed two weeks prior when he was admitted and treated for COPD exacerbation and cellulitis of the right shin. He converted to sinus rhythm after receiving 20mg IV diltiazem. Based on his CHA₂DS₂VASc score of 4, he was discharged home with 5mg twice a day (BID) of apixaban and metoprolol succinate 12.5mg BID for rate control; he was also started on amiodarone 200mg daily outpatient. A transthoracic echocardiogram before discharge revealed a normal ejection fraction of 62% with grade 2 diastolic dysfunction (Figure 1).

Figure 1. Parasternal long-axis view of the patient's baseline echocardiogram did not show pericardial effusion.

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On presentation to the ED, he was tachypneic with a respiratory rate of 32 breaths per minute and oxygen saturation of 91% on room air, his heart rate was 77 beats per minute and blood pressure was 114/78mmHg. Physical examination was remarkable for jugular venous distention, distant heart sounds, and 3+ bilateral lower extremity edema up to his shins. Abnormal laboratory findings on admission were hemoglobin of 10.8g/dL (his baseline is 12g/dL), creatinine of 1.67mg/dL (his baseline is 1.42mg/dL), glomerular filtration rate (GFR) of 40mL/min (his baseline is 50mL/min), blood urea nitrogen (BUN) of 31mg/dL, INR of 3.0, and N-terminal brain natriuretic peptide (NT-BNP) of 729pg/mL. Electrocardiogram revealed sinus rhythm with premature ventricular contraction, no low QRS voltage or electrical alternans (Figure 2). The chest radiograph was remarkable for an enlarged cardiac silhouette, new bibasilar airspace opacities, and small bilateral pleural effusion (Figure 3). A limited bedside echocardiogram was remarkable for a large pericardial effusion with fibrinous material noted in the effusion, an excessive respiratory variation of the mitral and

tricuspid inflow velocities, a dilated inferior vena cava (IVC) with a diameter >2.1cm, and <50% respiratory collapse estimating a right atrial pressure of 15mmHg. There was no collapse of the right ventricle (Figure 4).

Figure 2. Electrocardiogram unremarkable for low QRS voltage or electrical alternans.

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Figure 3. Chest radiograph remarkable for enlarged cardiac silhouette, new bibasilar airspace opacities, and small bilateral pleural effusion.

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Figure 4. Parasternal long-axis view on echocardiogram remarkable for a large pericardial effusion.

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He immediately received 50 units/kg of prothrombin complex concentrate to reverse the effect of apixaban and urgent pericardiocentesis was performed. There was 600mL of frank blood initially drained and an additional 200mL was drained by the end of the following day via the pericardial drain. The pericardial fluid was sent for cytology, acid fast bacilli (AFB) culture, stain, fungal culture, gram stain, and bacterial culture. A repeat echocardiogram a day after pericardiocentesis revealed complete evacuation of the effusion. Although the pericardiocentesis led to an improvement in his shortness of breath and orthopnea, he required diuresis with 20mg of furosemide for complete resolution. The pericardial drain was removed on day 3 of admission and he was discharged the following day.

Cytology of the pericardial fluid revealed numerous red blood cells, lymphocytes, neutrophils, and benign mesothelial cells. There were no malignant cells. There were no organisms on gram stain and bacterial culture was negative after five days. No organisms were noted on the AFB stain; AFB culture remained negative after five weeks. Fungal culture remained negative after four weeks.